

Glossary of Terms

COURSE 4 | DEVELOPING APPLICATIONS IN PYTHON ON AWS



Application Programming Concepts

Application programming interface (API): A set of rules and protocols that define how different software applications can communicate and interact with each other.

Array: A data structure that stores a fixed-size sequence of elements of the same type.

Asynchronous: Operations that occur independently and do not require immediate responses.

Boolean: A data type that represents either true or false values.

Conditional: A programming construct that executes code differently based on whether certain conditions are met.

Data type: The type of data that a variable can hold, such as integers, strings, or booleans.

Debug: The process of identifying and fixing errors or bugs in a program.

Dictionary: A data structure in Python that stores key-value pairs and allows quick access based on the keys.

Expressions: Combinations of values, variables, and operators that are evaluated to produce a result.

Floating point: A data type that represents numbers with decimal points.

Frameworks: A collection of libraries and tools that provide a foundation for building applications.

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Application Programming Concepts

Function: A reusable block of code that performs a specific task when called.

Integer: A data type that represents whole numbers without fractional parts.

Libraries: Collections of pre-written code that provide additional functionality to programs.

List: A data structure in Python that stores a collection of items.

Loop: A programming construct that repeats a set of instructions until a certain condition is met.

Map: An abstract data type that stores a collection of key-value pairs with unique keys.

Operations: Actions performed on data, such as arithmetic, comparison, or logical operations.

Operator: A symbol or keyword used to perform specific operations on data.

Packages: Bundles of related modules or libraries that can be easily installed and imported into a program.

Parameter: A variable used in a function to receive input values when the function is called.

Python interpreter: A program that reads and executes Python code, translating it into machine-readable instructions.

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Application Programming Concepts

Representational State Transfer

(REST): A software architectural style for designing networked applications, commonly used in web services development.

Return: The value that a function gives back after it completes its execution.

Runtime: The environment where code runs. It includes the necessary libraries, tools, and components that make it possible for the code to be interpreted, or compiled and run, on a specific platform or operating system.

Semantics: The meaning or interpretation of code in a programming language, which determines how it behaves.

String: A data type that represents a sequence of characters, such as text.

Structured Query Language (SQL):

A language used for managing and querying relational databases.

Synchronous: Operations that occur in real time and require immediate responses.

Syntax: The rules that define how code must be structured and written in a programming language.

Terminal: A text-based interface used to interact with a computer's operating system or execute commands.

Variable: A named storage location in a program that holds a value or data.

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Data Formats

Extensible Markup Language (XML):

A data format used for encoding documents into a machine and human readable format.

JavaScript Object Notation (JSON): A data interchange format that is easy for humans to read and write, and easy for machines to parse and generate.

YAML: A human-readable data serialization format often used for configuration files.

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AWS Services and Concepts

Amazon API Gateway: An AWS service that enables the creation, deployment, and management of APIs.

AWS CloudFormation: A service that you can use to define and provision AWS infrastructure as code.

Amazon EventBridge: A serverless event bus service provided by AWS.

Amazon DynamoDB: A fully managed NoSQL database service provided by AWS.

Amazon Relational Database Service (Amazon RDS): A managed database service provided by AWS.

Amazon RDS Proxy: A service that helps applications manage connections to relational databases.

Amazon Simple Notification Service (Amazon SNS): A fully managed pub-sub messaging service provided by AWS.

Amazon Simple Queue Service (Amazon SQS): A fully managed message queuing service provided by AWS.

AWS Lambda: An AWS serverless compute service that lets you run code without provisioning or managing servers.

AWS Lambda execution permissions: The permissions required for the code within a Lambda function to execute properly. They are provided through an IAM role.

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AWS Services and Concepts

AWS Lambda handler: The method in a Lambda function that processes events, or the entry point to your code within an AWS Lambda function.

AWS Lambda resource-based policy: An access control for AWS Lambda functions. This type of policy determines how principals can interact with an AWS Lambda function.

AWS Serverless Application Model (AWS SAM): An open-source framework for building serverless applications on AWS.

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Application Architectures

Microservice application architecture:

An architectural style that structures an application as a collection of loosely coupled services.

Monolithic application architecture:

An architectural style where an entire application is developed as a single unit.

Service-oriented application architecture:

An architectural style that organizes an application as a set of services that communicate over a network.

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Messaging Patterns

Event: A signal or notification indicating that something has occurred in a system.

Event buses: Communication channels that facilitate the exchange of events between different parts of a system.

Event-driven applications: Applications that use events as triggers and to communicate between decoupled services. They are common in modern applications built with microservices.

Point-to-point messaging: A messaging pattern where a message is sent from one sender to one specific receiver.

Publish-subscribe messaging:

A messaging pattern where a message is sent to multiple receivers who have expressed interest in, or subscribed to, that type of message.

Queue: An asynchronous communication mechanism between services that allows them to exchange messages or events with loose coupling.